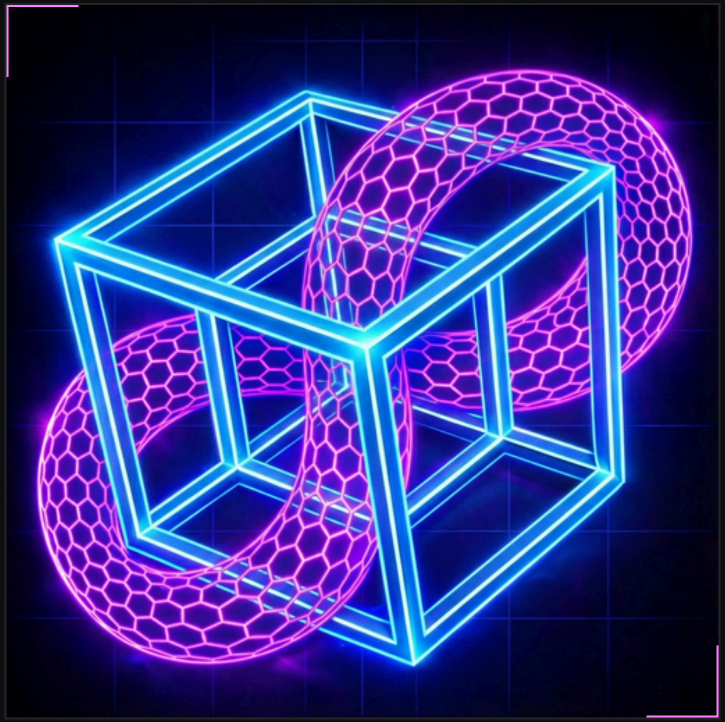


PolyStore

Decentralized Storage and Delivery

Decentralized storage has focused on keeping data.
PolyStore also delivers data.

mike@polynomialstore.com

THE PROBLEM

Stored Does Not Mean Delivered.

Decentralized storage networks can prove that data is stored, but reliable delivery remains weakly enforced.

Decentralized storage networks do not guarantee reliable retrieval.

THE SOLUTION

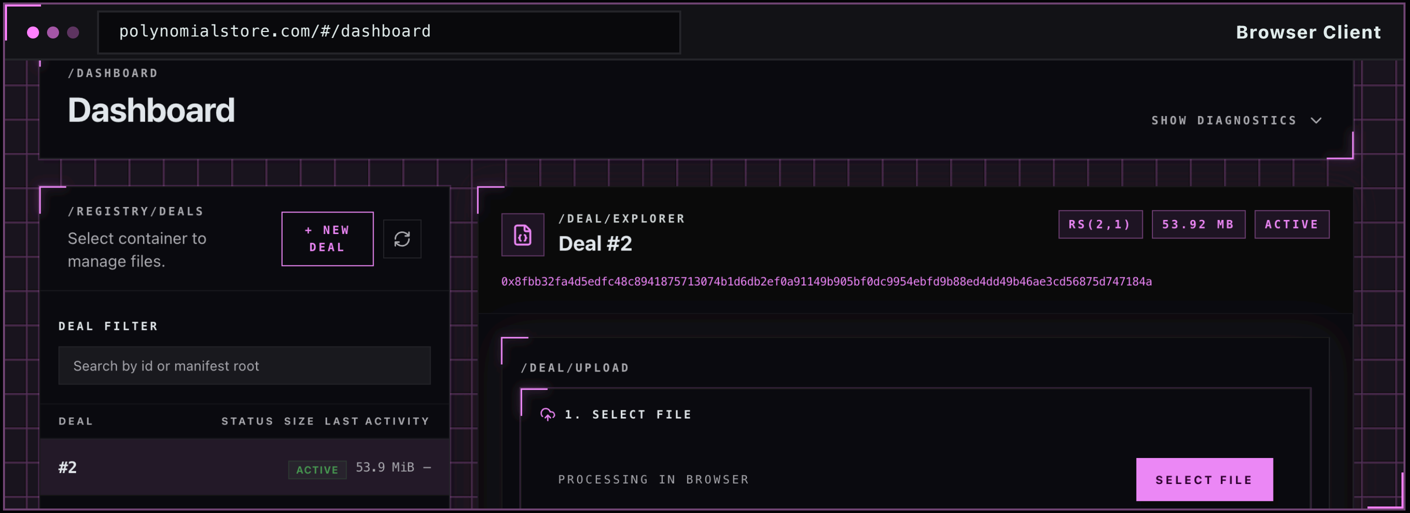
PolyStore Proves Delivery.

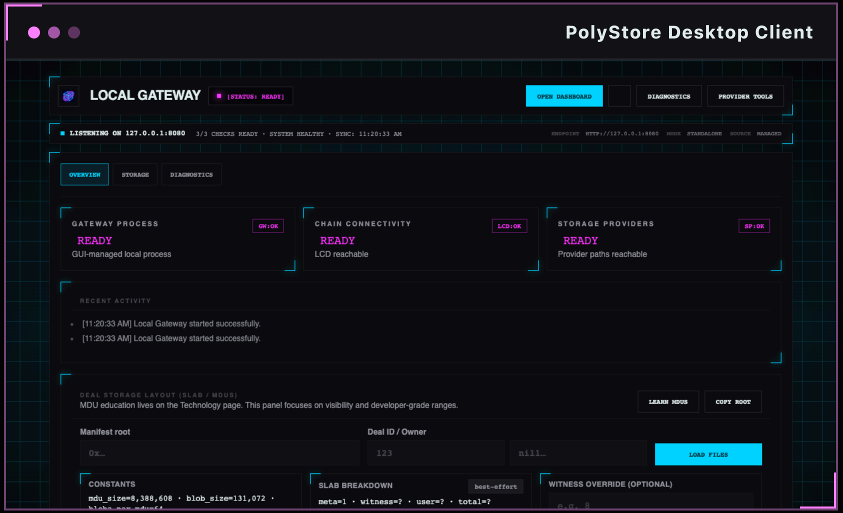
PolyStore uses efficient KZG Polynomial Proofs to cryptographically prove storage and retrievals on-chain.

PolyStore's protocol holds storage providers accountable for both storing and delivering data.

SOFTWARE TODAY

PolyStore Software Stack





>_ Terminal · CLI Client

```
$ nil-cli shard dataset.tar
Sharding file: "dataset.tar"
Processing MDU 1/4...
Computing Manifest...
PolyFS Root: 0x92f98d4f…
```

>_ Daemon · Protocol Node

```
$ polystorechaind start
CometBFT · block production active
RPC     127.0.0.1:26657
LCD     127.0.0.1:1317
EVM RPC 127.0.0.1:8545
```

>_ Daemon · Storage Provider

```
persona    provider-daemon
endpoint   127.0.0.1:8091
provider #1 registered
shard service ready
retrieval path healthy
```

</> Shared Core

- Rust
- PolyFS
- Erasure coding
- KZG commitments
- WASM + native

One shared core spans clients, protocol nodes, and storage providers.

NETWORK SCALE

PolyStore Scales

Testnet capacity is **~3.6 million** retrieval sessions per day. Planned optimizations to increase to **12 million**.

Channels can 1,000x capacity and support 12 billion.

2 Layers of recursive subchains with 30 chains each can 900x capacity and support 10 trillion deliveries per day

PolyStore enables On-Chain Storage and Retrieval at S3 Scale (17.28 Trillion requests per day) and Cloudflare Scale (11 Trillion requests per day)

CAPACITY MODEL

PolyStore Scales

BASE CHAIN

~3.6M retrieval sessions/day

Current testnet capacity
Planned: ~12M retrieval sessions/day

CHANNELS

~12B retrieval sessions/day

1,000× modeled capacity

SUBCHAINS

~10T deliveries/day

900× modeled parallel capacity
Two recursive layers

Modeled capacity approaches Cloudflare and Amazon S3 request volumes.

Founder / Protocol Lead

Mike Seiler

Cornell Applied Physics
BS · M.Eng.

Protocol Contributions

Filecoin EVM compatibility FIPs, Ethereum EIP-918.

Startup + Product Execution

Founding lead developer at Moat, acquired by Oracle in a reported $850M transaction. CTO and co-founder of Torch, a social media research and intelligence platform for government, political campaigns, and nonprofits.

MARKET ENTRY

Start With Decentralized Storage Users

INITIAL MARKET

Decentralized storage users and operators

Existing Filecoin, Arweave, IPFS, and Storj users already encountering failed retrievals.

NEXT

Protocol + appchain teams

Production datasets with real readback requirements.

LONG TERM

High-volume shared data

AI, media, archives, and public data systems.

Start with users already experiencing failed retrievals, then expand to broader production workloads.

CURRENT PRIORITY

Build the Early Community

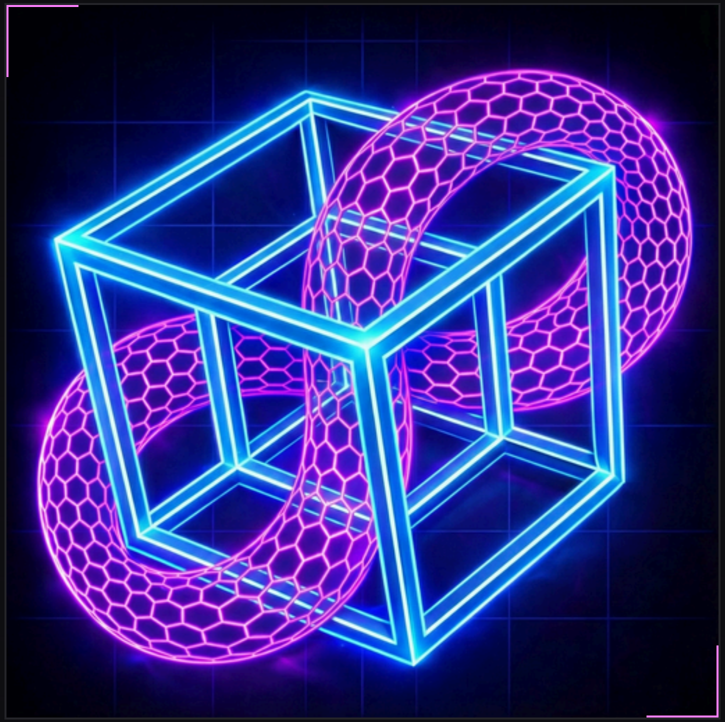
Workloads Real datasets with retrieval needs

Operators Providers and gateways running on the testnet

Team Prospective co-founders

Funding Aligned early investors

Early market development with the right people involved.



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APPENDIX · DEAL FLOW

How a PolyStore Deal Delivers Data

1 FUND THE DEAL

Reserve funding for storage and retrieval.

2 COMMIT THE DATA

PolyFS anchors the exact file state.

3 ASSIGN PROVIDERS

Place erasure-coded shards across provider slots.

4 SERVE + VERIFY

Serve requested bytes and verify them against the committed PolyFS root.

5 PAY OR REPAIR

Verified delivery earns payment. Repeated failure changes placement.

Fund storage and retrieval, commit the file state, verify delivery, then pay or repair.

PolyFS Primer

PolyFS anchors each Deal generation with a **32-byte root** committed on-chain. Off-chain proofs bind retrieved bytes to that state.

Blob
128 KiB
KZG commitment
48 bytes
BYTE-LEVEL COMMITMENT

→

MDU
64 blobs / 8 MiB
Merkle Data Unit root
32 bytes
EXACT FILE INCLUSION PROOFS

→

PolyFS
MDU #0 root table
Authenticates up to
65,536 MDUs
RETRIEVAL VERIFICATION

→

Deal
Up to 512 GiB-class
32-byte root
committed on-chain
DELIVERY EVIDENCE

One 32-byte root anchors the file state used to verify retrievals.

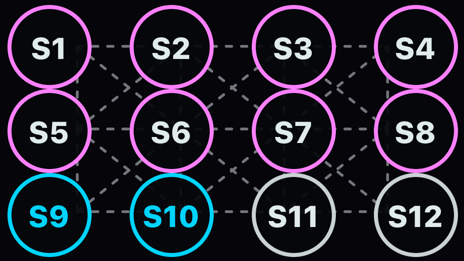
Verified Delivery Earns Payment

Repeated failure can lower reputation, result in penalties, and trigger repair.

FUNDED SESSION
BEFORE DELIVERY

- Payment is reserved.
- The session specifies which bytes to serve.
- One provider is responsible for serving them.

PROVIDER NETWORK
RS(8,12)



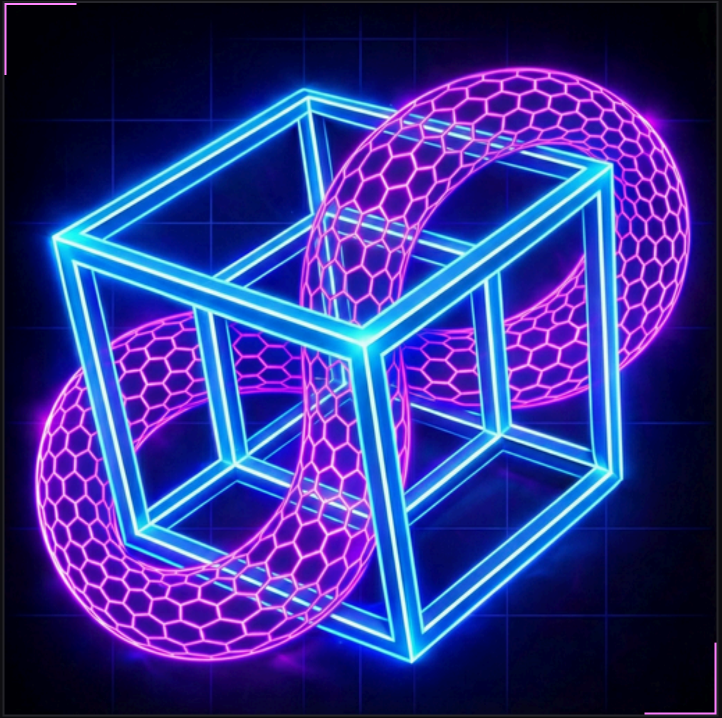
Any **8 healthy** slots can reconstruct the row.

MEASURED RESULT
AFTER DELIVERY

- Delivery verified against the PolyFS root earns payment.
- Failure can lower reputation or result in a penalty.
- Repeated failure triggers repair.

Healthy (8) Repair (2) Unavailable (2)

Payment, reputation, penalties, and repair follow measured results — not self-reported uptime or unverifiable claims of service.



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